Acoustic Anomaly Detection of Machinery using Autoencoder based Deep Learning

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***Abstract*—This research paper focuses on the domain of Acoustic Anomaly Detection (AAD) in industrial machinery using Deep Learning techniques. The primary objective of this study is to develop a reliable system for detecting anomalies in the acoustic patterns of industrial machines. In large-scale industrial environments, the early detection of faults and anomalies is critical to ensure the uninterrupted operation of machinery, minimize downtime, and optimize maintenance efforts. The datasets used in this research encompass a range of Signal-to-Noise Ratios (SNR) to simulate diverse operating conditions for industrial fans and pumps. After preprocessing each audio sample was transformed into 9 segments of shape 128 x 32 log mel-spectrograms. The study encompasses a comprehensive analysis of Convolutional Autoencoders (CAE) and Dense Autoencoders (DAE). These models are trained and evaluated against real-world industrial datasets (MIMII dataset), and their performance is meticulously compared. The results for the DAE and CAE had shown over 0.80 at the 6 dB SNR level and decaying results as the SNR level worsens. This research confirms some of the trends pointed out by the literature and provides detailed insight into how the autoencoders are developed and their properties could be used in order to detect anomalous behavior in audio data.**

***Index Terms*—Autoencoder, Acoustic Detection, Convolutional Autoencoder, Dense Autoencoder, ML, Unsupervised Anomaly Detection**

I. Introduction

In large-scale industrial environments, fault detection, anomaly detection, and predictive maintenance are of paramount importance. The industrial machinery in a plant is often expensive and requires regular maintenance and tuning to both minimize unforeseen breakdowns or failures during operation, and to extend the overall lifespan of the machines [1]. Undetected faults or anomalies can have detrimental effects on product quality control, leading to costly consequences such as defects in production, plant downtime or the need for machinery repairs or replacements. The complexity and variability of a machine's behaviour are both arbitrary and uncertain. Therefore, smart proactive measures are essential to narrow these uncertainties by detecting and addressing faults or anomalies before they escalate into serious and costly problems.

Classifying anomalies are notoriously difficult and are often machine or application specific, thus there is a high degree of variability in terms of mapping specific faults for each and every specific machine. This paper focuses on the adoption of Machine Learning (ML) using unsupervised deep learning techniques, such as autoencoder models, to develop an optimized data driven solution using a machine's operational acoustic data to classify anomalous behaviour.

This paper presents the related work in the field of ML in acoustics and anomaly detection in Section II. Section III explains the methodology used for the design of the ML models. The evaluation of results are explained in section IV and is followed by the conclusion in section V.

II. Related Works

A. Natural Language Processing

In recent ML studies, speech processing applications has emerged as a prominent area of interest. The primary focus of this research does not centre on speech processing, however it is integral to the development of the Acoustic Anomaly Detection (AAD) solution. The preprocessing and the convolutional neural networks commonly employed in speech processing applications demonstrate their robustness and applicability across diverse domains such as AAD. A novel speech emotion recognition model by Uddin et al [2], was based on the use of MFCC's (Mel-Frequency Cepstral Coefficients) as features, discriminant analysis which focuses on maximising inter-class scatterings and minimises inner class scatterings and a neural structured learning model. This combination had shown higher classification results than the other traditional models such as RNN, CNN and LSTM that used MFCCs.

D. Issa et al had used a deep CNN model to classify emotions based on speech data [3]. They had used three popular datasets in the field. Their multiclassification supervised approach had used a 1D CNN network that was fed by MFCCs, chroma-grams, mel-scale spectrogram, Tonnetz representation, and spectral contrast features. They had shown acceptable classification scores with the best being for the EMO-DB at 95.71% for 531 audio samples categorised into 7 classes for each emotion. There were other studies similar to these employing different variations in spectral, temporal and cepstral features, as well as various CNN model architectures [4], [5] and wavelet approaches [6].

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B. Autoencoders

Autoencoders differ from tradition neural networks as they are modelled to recreate the input image and minimize on the reconstruction error, which is the difference between the input and reconstructed image. Autoencoders are used widely for applications such as cyber security and network intrusion detection [7], credit card fraud detection [8] and leveraging data from various sensors for industrial predictive maintenance and quality control [9], [10]. There is a lot of research that is focused on the environmental results of sensors and not directly on the machine [10].

A study showed the development of an unsupervised autoencoder for AAD in an industrial setting by Duman et al [11]. They had uniquely created a custom dataset that extracted industrial sounds from YouTube videos which included painting, cutting, welding and robotic arm applications. They had also mixed different SNR (between 10 dB and -30 dB) to simulate noise to mimic an industrial environment as well as acoustic noises for generating explosions, fire, and glass-breaking events. OCSVM and autoencoder models were used with Mel-spectrograms in which features were extracted. They concluded that anomalies with distinct acoustic signatures such as the glass breaking events were much easier to detect. The results for the audio of the robotic arm performed poorly when compared with the other classes as they had recorded different variants of the robotic arm machine type. This shows motivation for the AAD to be a machine-type dependent algorithm.

CAE and DAE models were also utilized by Ribeiro et al [12], as part of the DCASE 2020 Task 2 challenge, which is a subtask of a broader challenge focusing on various acoustic events as opposed to specialised machine condition monitoring. The dataset used was a combination of the ToyADMOS and MIMII datasets which consisted of 6 machine types. They had achieved an AUC score of 72.03% and 73.06% for the fan and pump respectively.

C. Transfer Learning

Transfer learning is an approach commonly used in image classification and natural language processing. It leverages the trained parameters of pre-existing models in a similar field or domain and uses the weights in a custom model while adding additional layers and tuning existing hyperparameters [13]. Müller et al had proposed using 4 different CNN architectures (ResNet18, ResNet34, AlexNet and SqueezeNet) as feature extractors to find the features of Mel-spectrograms generated from the MIMMI dataset [14]. Slices of each audio clip were to be fed as feature vectors into the various anomaly detection models (GMM, B-GMM, Isolation Forest, KDE, OC-SVM and DCAE models were also created with one being based on a LeNet style model) that were tested. The results for the autoencoder configurations had shown comparable or weaker performance when compared with the transfer learning approaches. They had tested 4 different models of 4 different sets of machinery, the extremely varying result showed that the robustness of each ML model is dependent on the machine model (a variant of the machine) and machine type with the GMM and OC-SVM approach shown to have the best AUC readings for most of the configurations.

D. Data Augmentation

Salamon et al. had proposed a CNN architecture with data augmentations which is a common approach for image-based CNNs but not for audio-based ones [15]. This strategy is used in supervised image classification as it can create variation within a dataset and thereby increase the amount of data or parameters that help to generalise a deep neural network for better prediction rates. They had proposed to add data augmentations in the form of time-stretching, pitch shifting, dynamic range compression and additive background noise before converting them into spectrograms for feature extractions. In their research, they showed that most classes performed with better accuracy in their prediction when trained with augmented data however, there were some outliers that performed marginally worse such as an air conditioner, which performed dismally with the addition of background noise. Fans and pumps generally output the same type of cyclic noises, this suggests that data augmentation of this nature won't be a viable approach for the dataset that the AAD will implement.

III. Methodology

A. Data acquisition: MIMII Dataset

The study utilized the widely used MIMII dataset (Sound Dataset for Malfunctioning Industrial Machine Investigation and Inspection) by H. Purohit et al. [16], focusing on industrial fans and pumps. Each machine type had data from four models recorded with eight microphones in a circular array. Only one audio channel was used to reduce computational requirements. Normal operation and fault-induced sounds were recorded, such as changes in voltages, clogging, unbalanced fans for industrial fans, and leakage, contamination, and clogging for pumps [16]. Real factory sounds were mixed at -6 dB, 0 dB, and 6 dB noise levels. A custom dataset will be studied later on, as the MIMII dataset lacks information on specific fault labels to anomalies induced in the audio clips and the machinery used.

B. System Design and Technical Specifications

The system model is shown in Fig. 1. The data for each machine type and each model number is processed separately. For example, the ML model that uses the files from "fan_id_00" was modelled separately from fan_id_02 and so forth i.e. each model variant of each machine type will get its own ML model developed. The training data for the fan contained 650 normal samples whereas the test data contained a combination of 325 normal and 325 abnormal samples with a 50% ratio split to keep the testing stage balanced. The pump uses 100 normal and 100 abnormal samples for the test dataset and 600 samples for the training data as they are fewer recordings. The files were picked randomly using different seeds (42, 52, and 62)

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and each model was run 3 times and averaged to improve generalisation.

The ML model presented was developed using Python 3.9. The development process leveraged the capabilities of Google Colab and Microsoft Visual Studio Code, with experimentation conducted through Jupyter notebooks and Python scripts. The results presented here is a culmination 144 different models that were trained using Google Colab, which offers access to a high-performance Nvidia A100 GPU, significantly expediting the computational training procedures. TensorFlow's functional API served as the foundation for constructing these ML models. Audio and DSP tasks were accomplished using the Librosa library, while scikit-learn provided the necessary tools for statistical analysis of the results. Data management and control of randomization for file selection, crucial for mitigating overfitting, were facilitated by Pandas and NumPy.

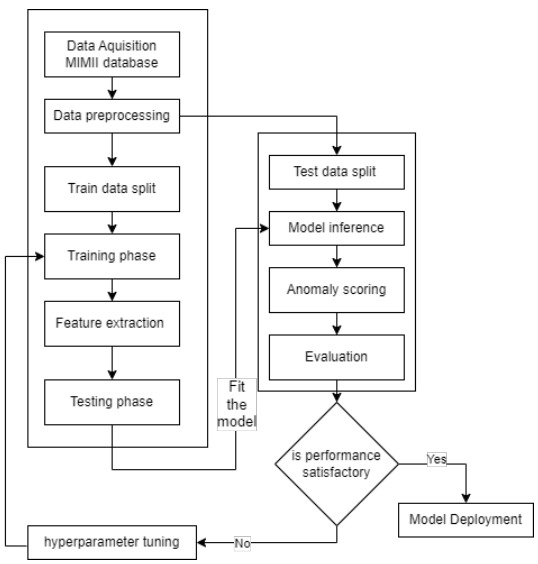

Fig. 1: Methodology

C. Preprocessing

Algorithm 1 shows the preprocessing pseudocode that was applied. The path of each file is loaded into memory and organised by use of a Pandas dataframe. The raw audio '.wav' files are loaded as a time series vector and sampled at 16 kHz shown by (1). A spectrogram is a visual representation of the spectrum of frequencies for a signal as it varies with time. It is shown best as a heatmap in a 2D plot of frequency and time, with the colour at a specific point indicating the intensity. This then creates the visual data that allows for CNN techniques to leverage into the classification models [17]. The Short-time Fourier Transform (STFT) shown in (2) is the fast fourier transform applied to a series of smaller-sized windows that are 1024 samples wide that sweep across the entire duration of the signal, which can be visualised in Fig. 2. The Hanning windowing technique is used to segment the signal and minimize artifacts introduced by abrupt signal transitions at the edges of the frames [18]. The stride at which

Algorithm 1 Preprocessing

```
Load Libraries
function PREPROCESS(filepath, segment_size = (128, 32), shift_size = 32)
    audio_path ← filepath
    audio, sr ← librosa.load(audio_path, sr=None)
    n_fft ← 1024
    hop_length ← 512
    n_mels ← 128
    spectrogram ← np.abs(librosa.stft(audio,
            n_fft=n_fft, hop_length=hop_length)
    mel_spectrogram ← librosa.feature.melspectrogram(
            S=spectrogram,
            sr=16000, n_fft=n_fft, hop_length=hop_length,
            n_mels=n_mels)
    segments ← []
    for i ← 0 to log_mel_spectrogram.shape[1] − segment_size[1] + 1 with step shift_size do
        segment ← log_mel_spectrogram[:, i : i + segment_size[1]]
        segments.append(segment)
    end for
    return log_mel_spectrogram, np.array(segments)
end function
```

the frames moves across the window is constrained by the hop length R which is equal to 512 samples in duration. The overlap of each consecutive frame helps to reduce spectral leakage that results from discontinuities when windowing. m is the time frame index, ω is the angular frequency, and w is the windowing function. N is the length of input signal $x(n)$. The window function $w(n)$ is centred at time mR before being applied to the Fourier transform.

$$X[n] = \{X[0], X[1], X[2], \ldots, X[159999]\} \tag{1}$$

$$X(m, \omega) = \sum_{n=0}^{N-1} x(n) \cdot w(n - mR) \cdot e^{-j\omega n} \tag{2}$$

To scale the spectrogram effectively, a logarithmic function is applied. The mel spectrogram is generated by applying a non-linear transformation to the spectrogram, where the frequency axis f, is converted from a linear scale to a Mel scale, the conversion is shown in (3) [17]. The resulting frequencies are mapped to the corresponding mel-frequency bins, of which 128 bins were selected. This approach is motivated by the sensitivity of human ears to different amplitudes as humans can perceive subtle variations in sounds with lower intensities more easily than with higher intensities. The algorithm then shows the segmentation of the resultant log mel-spectrogram into 9 segments with a tensor shape of 128 by 32, to be fed as input into the ML models.

$$M(f) = 2595 \log_{10}\left(1 + \frac{f}{700}\right) \tag{3}$$

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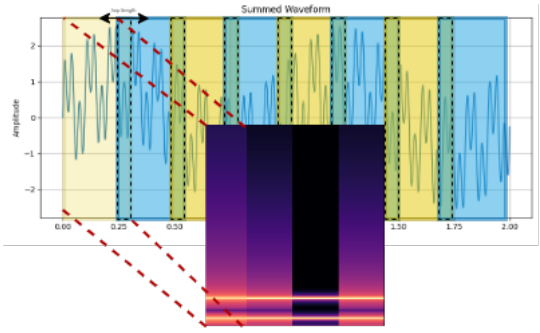

Fig. 2: Formation of a spectrogram using the STFT

D. Dense Autoencoder

The DAE was adapted and altered from the dataset provider [16]. This model consists of 3 Dense hidden layers of 512 neurons, each for the encoder and decoder portion. The DAE design for the baseline is as follows in Fig. 3 where n input image segment tensors of shape (128,32) are flattened into a column vector and fed into the autoencoder. Batch normalisation is used after each successive Dense layer as well as the Relu activation function. The latent dimension is made up of 8 neurons and is fed to the decoder network where the output is a 1-dimensional array and is reshaped to match the input shape. The Adam optimizer is a gradient descent-based optimization algorithm and was used with a learning rate of 0.01. The speed at which the model is tuned by the learning rate and takes effect during back propagation of the weights of each epoch by determining how big of a step to take during each iteration. In doing so, it controls the cost function which is the loss metric that was chosen to be the mean squared error (MSE) of the input and reconstructed image. The model was fitted with the normal training data (for the features and labels) and had a validation split of 0.1 and was ran for 15 epochs with a batch size of 32.

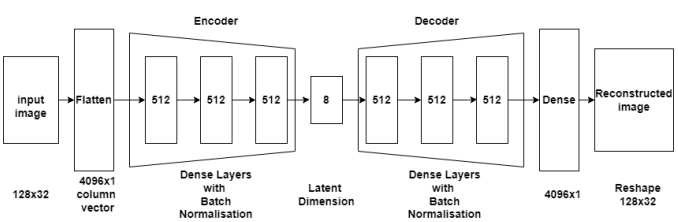

Fig. 3: Dense autoencoder architecture

E. Convolutional Autoencoder

The second model architecture is a custom convolutional autoencoder and was inspired the work done by Müller et al. [14] and Ribeiro et al. [12]. It is shown in Fig. 4. It takes in n input image segment tensors of shape (128, 32, 1) which corresponds to the image height, width, and number of channels respectively. The encoder and decoder portion are made of 3 convolutional layers with each containing 256, 128 and 64 convolutional filters with a kernel filter of (5, 5). The first two and last two layers had used a stride of (1, 2). The inner layers used a stride of (2, 2). The latent space of 32 filters used a stride of (1,1) and a kernel size of (4, 4). The Adam optimizer was used with a learning rate of 0.001 and validation split of 0.1. The decoder uses 2D Convolutional Transposed layers which is used to build up the spatial dimensions from the encoded data. The loss function optimizes the MSE metric during each epoch. Early stopping was used to stop training as the loss metric converges if there was no improvement. Batch normalisation is used to speed up optimization resulting in a stable and faster computation.

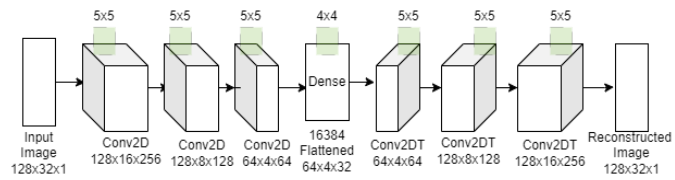

Fig. 4: Convolutional autoencoder architecture

F. Anomaly Scoring and Thresholding

After training the model on normal data, the model is then used to make inferences on the test dataset. The reconstruction error (MSE) in (4), is computed for each segment. The 9 segment errors were then averaged. It was debated on using the maximum error of the segments instead, but the average errors yielded better results overall. The resulting error array is combined with ground truth labels for the ROC (Receiver Operating Characteristic curve) analysis. The ROC analysis calculates TPR (True Positive Rate) and FPR (False Positive Rate) for various thresholds. Youden's index, J in (5), helps to find the optimal threshold, which is a common approach in many ML applications [11], [19]. Errors above this threshold are considered anomalous. After the best threshold was determined, classification metrics could then be calculated to evaluate model's performance in a standardised rigor.

$$\text{MSE} = \frac{1}{n}\sum_{i=1}^{n}(x_i - \hat{x}_i)^2 \tag{4}$$

$$J = \text{Sensitivity} + \text{Specificity} - 1 \tag{5}$$

IV. Evaluation of Results

Table. I presents the averaged list of results for all machine runs. The first noticeable trend was the significant improvement in the AUC (Area Under the Curve) reading and the F1-score when moving from a worse -6 dB SNR level to a better 6 dB SNR level. The DAE was able to improve its AUC score for the overall fan models from 0.643 to 0.919, as most of the individual models had achieved a score over 90 percent. The overall pump score had also shown the same behaviour with a score from 0.659 at -6 dB to a respectable 0.847 AUC score at 6 dB, which was ultimately lowered by pump model 02 performing poorly at 0.505. The Fan id_06 and Pump id_04 model results shown in Fig. 5 through Fig.

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7 and Fig. 8 through Fig. 10 respectively, had shown to have the best response to the DAE models applied to the data sets with the pump having near perfect AUC scores at the 6 dB SNR level. The confusion matrices and error curves provide visual feedback on the model's classification ability to predict true positives and true negatives along the diagonals. The CAE score had performed comparatively the same if not worse than its DAE counterpart.

The dataset provider had provided their own AUC scores for their models [16]. The models presented here were comparable to their results with some overall scores beating their own. Their overall average pump AUC score at 6 dB was a reading of 0.81 which is comparable to the CAE model and was beaten by the DAE model which scored an overall of 0.847. The DAE score for the pump at 0 dB scored 0.777 and had beaten their score which was 0.74. It was expected that the CAE would be able to find more unique trends amongst the features as it uses a more complicated architecture that is computationally more expensive than the DAE. However, the results had shown a similar or worse performance. This could be attributed to the errors collected not being exacerbated enough when parsing anomalous data through the model i.e. the model generalised the data as normal.

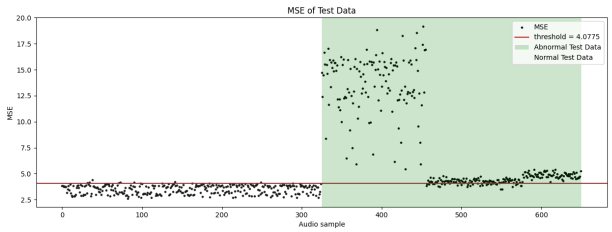

Fig. 5: Errors for samples - fan_id_02 6 dB seed 42

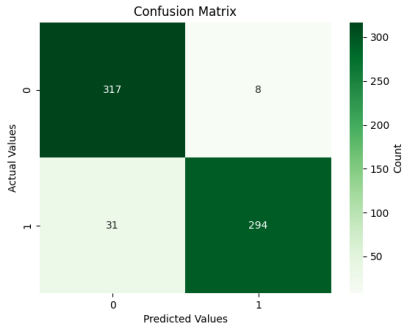

Fig. 6: Confusion Matrix - fan_id_02 6 dB seed 42

V. Conclusion

In this paper the aim was to use acoustic information generated by industrial machinery to detect anomalies using autoencoders (as a specific ML algorithm). This could be step towards predictive maintenance and quality control. The study employed the MIMII dataset, which includes data from industrial fans and pumps, and tested two different autoencoder architectures across various SNR levels. The findings revealed that the DAE performed the best in anomaly detection at the 6 dB SNR level whilst the CAE had performed similarly or

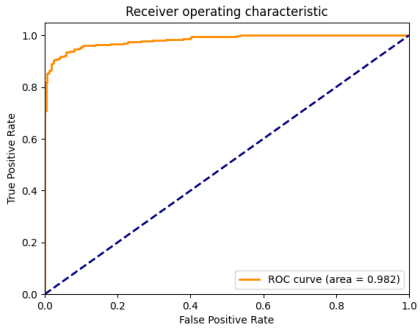

Fig. 7: Curve for ROC-AUC - fan_id_02 6 dB seed 42

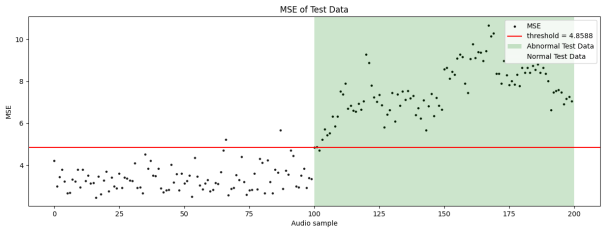

Fig. 8: Errors for samples - pump_id_04 6 dB seed 52

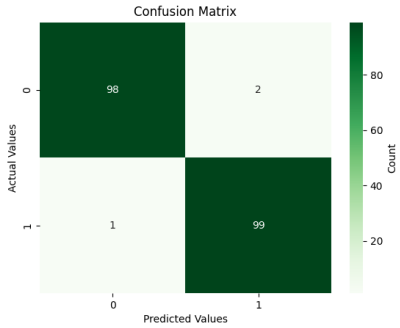

Fig. 9: Confusion Matrix - pump_id_04 6 dB seed 52

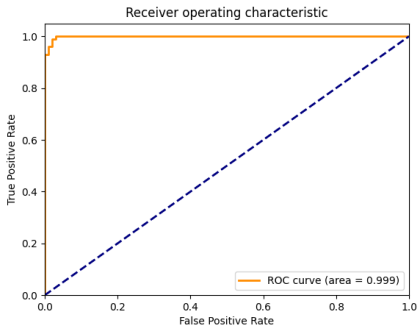

Fig. 10: Curve for ROC-AUC - pump_id_04 6 dB seed 52

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TABLE I: Results for DAE (left) and CAE (right)

SNR	Machine ID	Fan ROC-AUC		Fan Accuracy		Fan F1-Score		Pump ROC-AUC		Pump Accuracy		Pump F1-Score	
-6 dB	00	0.547	0.498	0.556	0.529	0.477	0.361	0.557	0.637	0.638	0.727	0.642	0.775
	02	0.682	0.678	0.635	0.683	0.666	0.572	0.554	0.404	0.593	0.542	0.432	0.602
	04	0.513	0.505	0.533	0.554	0.566	0.445	0.864	0.439	0.817	0.657	0.845	0.364
	06	0.785	0.717	0.737	0.733	0.749	0.619	0.661	0.441	0.645	0.550	0.546	0.348
		0.632	0.600	0.615	0.625	0.614	0.499	0.659	0.480	0.673	0.619	0.616	0.522
0 dB	00	0.577	0.595	0.581	0.593	0.598	0.450	0.893	0.775	0.848	0.792	0.840	0.833
	02	0.825	0.721	0.753	0.724	0.741	0.649	0.549	0.434	0.622	0.635	0.461	0.425
	04	0.765	0.552	0.723	0.587	0.729	0.332	0.898	0.993	0.847	0.977	0.860	0.977
	06	0.896	0.873	0.875	0.846	0.863	0.824	0.766	0.521	0.730	0.662	0.735	0.652
		0.766	0.685	0.733	0.687	0.733	0.564	0.777	0.681	0.762	0.766	0.724	0.722
6 dB	00	0.791	0.592	0.730	0.704	0.742	0.544	0.921	0.841	0.893	0.820	0.888	0.777
	02	0.947	0.766	0.902	0.813	0.890	0.766	0.505	0.505	0.620	0.648	0.406	0.460
	04	0.978	0.647	0.935	0.719	0.935	0.632	0.992	0.998	0.972	0.988	0.971	0.988
	06	0.959	0.828	0.908	0.858	0.899	0.837	0.969	0.870	0.940	0.868	0.937	0.808
		0.919	0.708	0.869	0.774	0.867	0.695	0.847	0.804	0.856	0.831	0.800	0.758

poorly in comparison. Further development will involve using more complex AE architectures on self obtained datasets and the deployment of such on embedded ML capable devices.

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